

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

UNIT 2-ASSIGNMENT I

Course Code: ITA0612

Course Title: Machine Learning

Section / Batch: Slot A

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Data: 22-07-2026

Explain the Role of the Error (Delta) Term in Backpropagation Using an Example of a Network Trained to Classify Malnutrition Risk in Children Under Five (SDG 2 – Zero Hunger)

Introduction

Backpropagation is a supervised learning algorithm used to train Artificial Neural Networks (ANNs). During training, the network predicts an output, compares it with the actual output, computes the prediction error, and updates the weights to improve future predictions. The **error (delta) term** is the key component that determines how much each weight should change.

In **SDG 2 – Zero Hunger**, machine learning models can help healthcare workers identify children under five who are at risk of malnutrition using features such as age, weight, height, BMI, and Mid-Upper Arm Circumference (MUAC).

Artificial Neural Network for Malnutrition Classification

A neural network consists of three layers:

- **Input Layer**
- **Hidden Layer**
- **Output Layer**

For malnutrition prediction, the input layer receives health-related parameters of a child.

Input Features

- Age
- Weight
- Height
- Body Mass Index (BMI)
- Mid Upper Arm Circumference (MUAC)

These features are processed by the hidden layer, and the output layer predicts whether the child is:

- Healthy (0)
- Malnourished (1)

During training, the network compares the predicted output with the actual output and computes the error. This error is then propagated backward through the network to update the weights.

What is the Error (Delta) Term?

The **delta (δ) term** is an error signal generated during the backpropagation process. It indicates how much each neuron contributes to the overall prediction error.

The delta value helps determine:

- Which weights should be updated.
- How much each weight should change.
- The direction of weight adjustment.
- How the prediction error can be minimized.

If the prediction error is large, the delta value will also be large, resulting in significant weight updates. If the prediction error is small, only minor weight adjustments are required.

The delta term ensures that the neural network gradually learns the correct relationship between inputs and outputs.

Role of the Error (Delta) Term

The **delta (δ) term** represents the error associated with a neuron. It is calculated by comparing the network's predicted output with the actual target output.

For the output layer,

$$\delta = (\text{Target} - \text{Output}) \times \text{Output} \times (1 - \text{Output})$$

where

- **Target** = Actual class
- **Output** = Predicted value
- **Output $\times$ (1 – Output)** = Derivative of the sigmoid activation function

The delta term is propagated backward through the network and is used to update the weights.

Weight update equation:

$$\text{Weight}_{\text{new}} = \text{Weight}_{\text{old}} + \text{Learning Rate} \times \text{Delta} \times \text{Input}$$

Example: Malnutrition Risk Classification

Suppose a neural network receives the following information about a child:

Feature Value

Feature	Value
Age	3 years
Weight	9 kg
Height	85cm
BMI	12.8
MUAC	11cm

Actual Class = **Malnourished (1)**

Neural Network Prediction = **0.72**

Step 1: Compute Error

Error = Target – Output

= 1 – 0.72

= **0.28**

Step 2: Compute Delta

Delta

= $0.28 \times 0.72 \times (1 - 0.72)$

= $0.28 \times 0.72 \times 0.28$

= **0.0564**

This delta value is propagated backward through the hidden layer to update the weights.

After several training iterations, the prediction gradually becomes:

- Epoch 1 → 0.72
- Epoch 2 → 0.81
- Epoch 3 → 0.89
- Epoch 4 → 0.94
- Epoch 5 → 0.98

The prediction becomes closer to the actual value, showing that the network has learned successfully.

```
import numpy as np
import matplotlib.pyplot as plt

# Target output (Actual)
target = 1

# Predicted outputs over training epochs
predictions = [0.30, 0.45, 0.60, 0.72, 0.82, 0.90, 0.96]

errors = []
deltas = []

for output in predictions:
    error = target - output
    delta = error * output * (1 - output)

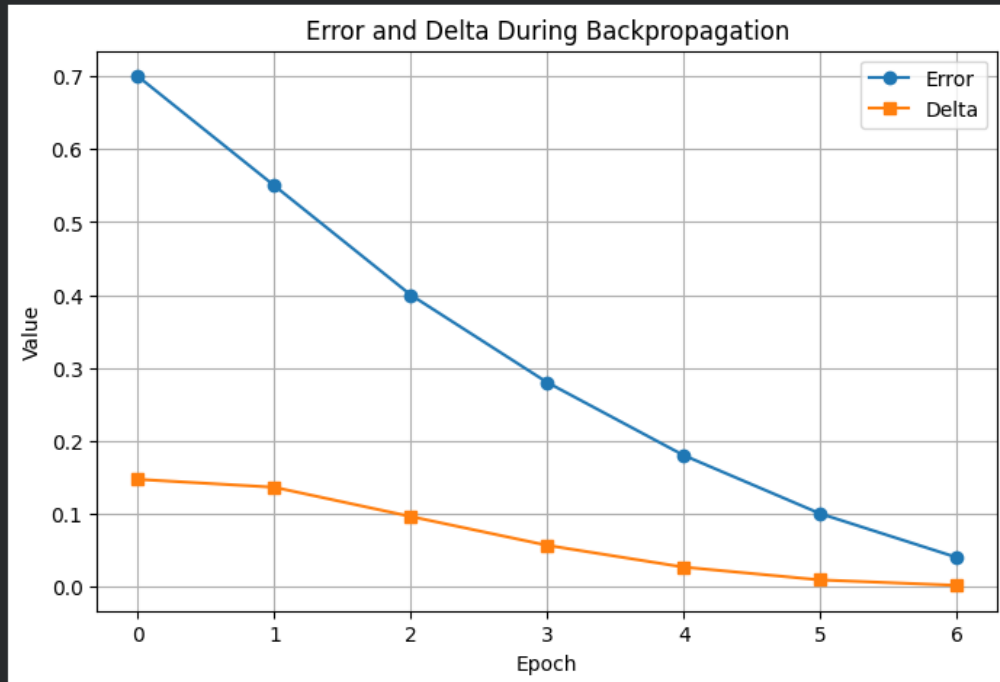
    errors.append(abs(error))
    deltas.append(delta)

print("Epoch\tPrediction\tError\tDelta")
for i in range(len(predictions)):
    print(i+1, "\t",
          round(predictions[i],2), "\t\t",
          round(errors[i],3), "\t",
          round(deltas[i],4))

# Plot Error and Delta
plt.figure(figsize=(8,5))
plt.plot(errors, marker='o', label='Error')
plt.plot(deltas, marker='s', label='Delta')
plt.xlabel("Epoch")
plt.ylabel("Value")
plt.title("Error and Delta During Backpropagation")
plt.legend()
plt.grid(True)
plt.show()
```

Output

```
... Epoch Prediction Error Delta
1 0.3 0.7 0.147
2 0.45 0.55 0.1361
3 0.6 0.4 0.096
4 0.72 0.28 0.0564
5 0.82 0.18 0.0266
6 0.9 0.1 0.009
7 0.96 0.04 0.0015
```



Role of the Delta Term in Backpropagation

The delta term performs several important functions during neural network training.

1. Calculates Prediction Error

The delta term measures the difference between the predicted value and the actual target value.

2. Updates Network Weights

Using the delta value, the neural network updates all weights to improve future predictions.

3. Reduces Overall Loss

As the delta becomes smaller, the prediction error decreases, leading to lower training loss.

4. Improves Classification Accuracy

Repeated weight updates enable the network to classify healthy and malnourished children more accurately.

5. Enables Learning

Without the delta term, the network would never learn from previous mistakes.

6. Propagates Error to Hidden Layers

The delta term sends the error information backward from the output layer to the hidden layers.

7. Supports Faster Convergence

Proper delta calculations help the network converge faster during training.

8. Prevents Large Prediction Errors

By continuously adjusting weights, the delta term minimizes future prediction errors.

Application to SDG 2 – Zero Hunger

Malnutrition is one of the leading causes of illness and mortality among children under five years of age. Machine Learning models trained using backpropagation can assist healthcare professionals by automatically identifying children who require nutritional intervention.

The neural network can analyze children's health records and classify their nutritional status with high accuracy. Such systems can be integrated into hospitals, rural health centers, and mobile health applications to provide early warnings about malnutrition. Early diagnosis enables governments and healthcare organizations to provide food supplements, nutritional counseling, and medical treatment before the condition becomes severe.

Thus, Artificial Neural Networks directly contribute to **Sustainable Development Goal (SDG) 2 – Zero Hunger** by improving child nutrition, reducing malnutrition-related diseases, and supporting better healthcare planning.

Advantages of Using Backpropagation

- Produces high classification accuracy.
- Learns complex relationships among health parameters.
- Automatically updates weights using the delta term.
- Suitable for large healthcare datasets.
- Reduces prediction error over time.
- Improves decision-making in medical diagnosis.
- Supports early identification of malnutrition.
- Can be integrated with healthcare management systems.

Limitations

- Requires a sufficient amount of training data.
- Training can take longer for large datasets.
- Performance depends on selecting appropriate learning rates.
- May suffer from overfitting if not properly regularized.
- Choosing the number of hidden neurons can be challenging.

Conclusion

The **error (delta) term** is the most important component of the backpropagation algorithm because it determines how much the neural network should adjust its weights after each prediction. By calculating the difference between the predicted and actual outputs, the delta term enables the network to learn from its mistakes and continuously improve its performance. In the malnutrition risk classification example, the neural network uses children's health parameters such as age, weight, height, BMI, and MUAC to identify whether a child is healthy or malnourished. As the training process continues, the delta term becomes smaller, indicating reduced prediction error and improved classification accuracy. This makes backpropagation an effective technique for healthcare applications and supports **SDG 2 – Zero Hunger** by enabling early detection of malnutrition, timely nutritional intervention, and improved child health outcomes. This combination of Machine Learning and healthcare contributes to building healthier communities and reducing child malnutrition worldwide.

